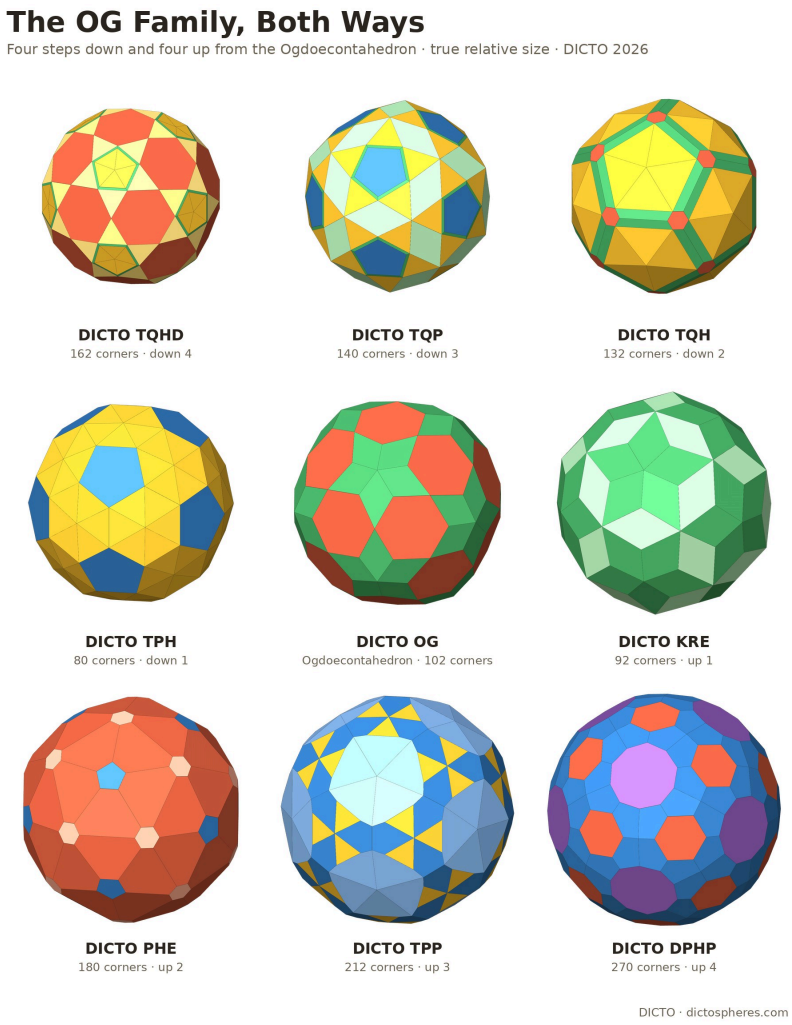
The DICTO OG Family: Four Steps Up and Four Down



Eight new icosahedral solids grown outward and inward from the DICTO Ogdoecontahedron by one rule and its reverse, exact in $\mathbb{Q}(\sqrt{5})$, with the shells between them and where the nesting breaks

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Paper #39 · Version 2 · 7 November 2026 (Version 1: 20 October 2026) · DICTO DOI: 10.5281/zenodo.23488944 (concept) · Companion to The DICTO Ogdoecontahedron (DICTO DOI: 10.5281/zenodo.23404688) and Universal PiTet: The DICTO PiTet Proposition (DICTO DOI: 10.5281/zenodo.23463305) · Licence CC BY-SA 4.0 · Scripts, exact checks and data: https://github.com/DICTOR-Master/kaleidohedra/tree/058c2fb/studies/2026-10-19-five-fold-octet (research branch, commit 058c2fb; also attached to this record); Version 2: https://github.com/DICTOR-Master/kaleidohedra/tree/research/studies/2026-11-04-nest-fingerprint and 2026-11-04-dehn-screen (research branch)
The idea of running the Universal PiTet step outward from the Ogdoecontahedron (“the upward shapes”), of finding the lower solids as well, and of one paper for both directions is DICTO's; the eight names are DICTO's (the four downward names in his naming scheme, approved by him). DICTO proposed the harmonic fingerprint of the golden nests and the OG family added in Version 2. Computation, checking and drafting with Claude (Anthropic).

Version 2 adds §5, Harmonic fingerprint and Dehn invariant: the nine solids' fingerprints (Proposition 1) and their nonzero Dehn invariants (Proposition 2), with Table 4.



Abstract. In Universal PiTet a single rule T builds each solid from the last: every corner v of a solid S gives the face plane v·x = v·v, through v and at right angles to the line from the centre, and T(S) is the solid these planes bound. Started at the icosidodecahedron it closes into a four-step cycle. DICTO proposed running it from the DICTO Ogdoecontahedron (OG), and also running it the other way. The reverse step P takes as new corners the foot points of S's face planes, the points of those planes nearest the centre. From the OG, four steps of T give four solids with 92, 180, 212 and 270 corners, the **DICTO KRE, PHE, TPP and DPHP**, and four steps of P give four with 80, 132, 140 and 162 corners, the **DICTO TPH, TQH, TQP and TQHD**. All eight are proved exactly in $\mathbb{Q}(\sqrt{5})$, each with the full icosahedral symmetry, and each step of P is undone exactly by T. Several have regular faces: 12 regular pentagons (PHE, TPH, TQP), 20 regular hexagons (PHE, DPHP, TQH, TQHD); the KRE has 30 rhombi and 60 kites. The family does not cycle. It nests as shells for three steps up, OG ⊂ KRE ⊂ PHE ⊂ TPP, and two steps down, TQH ⊂ TPH ⊂ OG, and each of these shells, and also the shell TQHD ⊂ TQP, is filled exactly by one piece per face of the inner solid. Both directions then break in the same way: 60 corners of the TPP stand outside the DPHP, and 60 corners of the TQP outside the TQH. In the plane, T and P are the classical antipedal and pedal constructions; the polyhedral pedal construction was posed as an open problem in 2010. We found the eight solids in no source. Version 2 adds the harmonic fingerprint and the Dehn invariant of all nine solids (§5): all have $Q_4 = Q_8 = 0$ and $\hat{W}_6 = \pm 0.16975$ (exact); the sign splits them into four of corner type (TQHD, TQP, OG, KRE) and five of face type (TQH, TPH, PHE,

> TPP, DPHP), changing in pairs along the chain, with the KRE the quietest (numerical); and all nine have a nonzero Dehn invariant (exact for the OG, numerical for the other eight), so none fills space alone.

1. The rule and its reverse

Frame. As in [2]: the icosidodecahedron of circumradius φ, and the OG between it and the rhombic triacontahedron, ID ⊂ OG ⊂ RT. In this frame the OG has 102 corners (radii φ and ≈ 1.6873), 80 faces (60 kites and 20 regular hexagons) and volume **(525 + 185√5)/57 ≈ 16.4679**. All coordinates are pairs of rationals a + b√5 with exact signs.

The step up, T [2]: T(S) = {x : v·x ≤ v·v for every corner v of S}. **The step down, P**: if S has the face planes n·x = d (the centre inside, d > 0), P(S) is the convex hull of their foot points (d/n·n) n. The foot point of a face plane is the corner of P(S) whose own T-plane is that face plane, so:

- **T undoes P** when every foot point is a corner of P(S): then T(P(S)) is bounded by exactly the face planes of S, so T(P(S)) = S. For all four downward steps this holds, and T(P(S)) = S is checked exactly.
- **P does not always undo T.** A corner of S whose plane is not a face of T(S) is lost. From the OG, the planes of its 12 five-fold corners touch the KRE only at those corners themselves (they become the KRE's 12 five-fold corners), so the KRE has 102 − 12 = 90 faces, and P(KRE) is the hull of the OG's other 90 corners, not the OG (exact).

Proof method (as in [2]). Floating point only guesses which faces meet at a corner; each guess is proved exactly: corners solve their faces and lie inside every face; each face holds at least three corners with the others inside it; every edge lies in exactly two faces and V − E + F = 2. Symmetry is checked with the 60 exact rotations of the icosahedral group, faces are compared by exact side lengths and corner angles, and volumes are exact sums of determinants.

2. Four steps up (exact)

Solid	Corners	Faces	Edges	Volume	Corner radii
DICTO OG	102	60 kites + 20 regular hexagons (80)	180	(525 + 185√5)/57 ≈ 16.4679	1.6180–1.6873
DICTO KRE, Kite-Rhombic Enneacontahedron	92	30 rhombi + 60 kites (90)	180	(920 + 60√5)/57 ≈ 18.4941	1.6520–1.7321
DICTO PHE, Pentagon-Hexagon Enneacontadihedron	180	12 regular pentagons + 20 regular hexagons + 60 hexagons (92)	270	(95875 − 1560√5)/4617 ≈ 20.0101	1.7046–1.7427
DICTO TPP, Triangle-Pentagon Polyhedron	212	60 triangles + 120 pentagons in two kinds (180)	390	≈ 22.3366	1.7220–1.8104
DICTO DPHP, Decagon-Pentagon-Hexagon Polyhedron	270	12 equal-sided decagons + 20 regular hexagons + 120 pentagons (152)	420	≈ 23.5434	1.7995–1.8095

Table 1. The volumes of the TPP and DPHP are exact but long; they are in data/exact_upward.json. Radii are float values of exact corners.

The KRE. Its 92 corners and 90 four-sided faces are the counts of the rhombic enneacontahedron (§5), but its faces are 30 rhombi (side² 29/18 − √5/2, side ≈ 0.7022) and 60

kites, each with two sides equal to the rhombus side and two of side² (1505 + 315√5)/6498 (side ≈ 0.5831). **The PHE** has the face counts of the Goldberg polyhedron GP(3,0), 12 pentagons and 80 hexagons [6, 7]; here the 12 pentagons (side² 35/18 − 5√5/6) and 20 of the hexagons (side² 1/27, side √3/9) are regular, the other 60 hexagons not. **The TPP** has isosceles triangles and two kinds of pentagons. **The DPHP** is the roundest: its corner radii differ by only 0.56%. Its decagons have ten equal sides but two alternating angles, so they are not regular (exact).

3. Four steps down (exact)

Solid	Corners	Faces	Edges	Volume	Corner radii
DICTO TPH, Triangle-Pentagon Hecatontatriacontadihedron	80	120 isosceles triangles + 12 regular pentagons (132)	210	(10437930 + 3959694√5)/109³ ≈ 14.8970	1.5115–1.5848
DICTO TQH, Triangle-Quadrilateral-Hexagon Polyhedron	132	60 triangles + 60 quadrilaterals + 20 regular hexagons (140)	270	(770805045 + 337172409√5)/(2⁴·41³·109) ≈ 12.6853	1.4885–1.5023
DICTO TQP, Triangle-Quadrilateral-Pentagon Polyhedron	140	60 triangles + 30 rhombi + 60 quadrilaterals + 12 regular pentagons (162)	300	≈ 11.6524	1.4172–1.4931
DICTO TQHD, Triangle-Quadrilateral-Hexagon Diacosihedron	162	120 triangles in two kinds + 60 quadrilaterals + 20 regular hexagons (200)	360	≈ 10.4208	1.3493–1.3948

Table 2. Exact volumes of the TQP and TQHD: data/exact_downward.json.

Going down, every foot point is a corner, so each solid has as many corners as the one above has faces: 80, 132, 140, 162. Going up, the corner count of the solid below is an upper bound for the faces of the one above, met at the PHE and TPP steps; the OG loses 12 (above) and the TPP 60 (§4). The downward solids leave the small golden numbers of the OG frame behind: the first step brings the prime 109 into the denominators (the TPH's pentagon side² is (4014 + 378√5)/109²), the second the prime 41. Every triangle in the family is isosceles; the regular faces are those named in the tables (exact).

4. Shells and where the nesting breaks (exact)

When one solid lies inside the next, the shell between them is filled by one piece per face of the inner solid: the cone from the centre over that face, cut off by the outer solid [2]. All eight neighbouring pairs were checked; six nest:

Shell	Pieces	Piece volumes	Gap
OG ⊂ KRE	60 on kites, 20 on hexagons	(230 + 15√5)/37278 ≈ 0.00707; (429 − 145√5)/1308 ≈ 0.08010	≈ 2.02617
KRE ⊂ PHE	30 on rhombi, 60 on kites	235/2916 − √5/36 ≈ 0.01848; ≈ 0.01603	≈ 1.51602
PHE ⊂ TPP	12 on pentagons, 60 on hexagons, 20 on regular hexagons	≈ 0.00218; ≈ 0.03805; 409/476766 ≈ 0.000858	≈ 2.32644
TPP, DPHP	**not nested:** 60 corners of the TPP lie outside the DPHP, by up to 0.0185		
TPH ⊂ OG	120 on triangles (two mirror orbits), 12 on pentagons	≈ 0.006748; ≈ 0.06343	≈ 1.57092
TQH ⊂ TPH	60 on triangles, 60 on quadrilaterals, 20 on hexagons	≈ 0.02937; ≈ 0.007311; ≈ 0.000550	≈ 2.21176
TQP, TQH	**not nested:** 60 corners of the TQP lie outside the TQH, by up to 0.0058		
TQHD ⊂ TQP	60 + 60 on the two kinds of triangle, 60 on quadrilaterals, 20 on hexagons	≈ 0.000016 and 0.003318; ≈ 0.000090; ≈ 0.05131	≈ 1.23162

Table 3. In every nested shell the pieces sum to the gap with remainder exactly 0, and pieces on congruent faces are congruent (one volume per symmetry orbit). Exact volumes not printed here are in data/exact_upward.json and data/exact_downward.json.

Why it breaks. Upward, T(S) contains S only if no corner w of S lies beyond another corner's plane, w·v ≤ v·v. On the TPP, whose corners lie at four radii and close together in direction, 60 corners fail this; exactly these 60 are the corners whose planes give no face of the DPHP, so the DPHP has 212 − 60 = 152 faces (exact). Going from the OG to the KRE no corner fails: the 12 lost there are the OG's five-fold corners, which become corners of the KRE. Downward, P(S) lies inside S only if every foot point falls inside its own face; 60 of the TQH's 140 face planes have their foot point outside the face, and those feet are the TQP's 60 outlying corners. So **the OG family nests exactly only near the OG**, OG ⊂ KRE ⊂ PHE ⊂ TPP upward and TQH ⊂ TPH ⊂ OG downward, unlike the UP cycle, whose solids nest for ever [2]. Below the break the nesting returns once (TQHD ⊂ TQP).

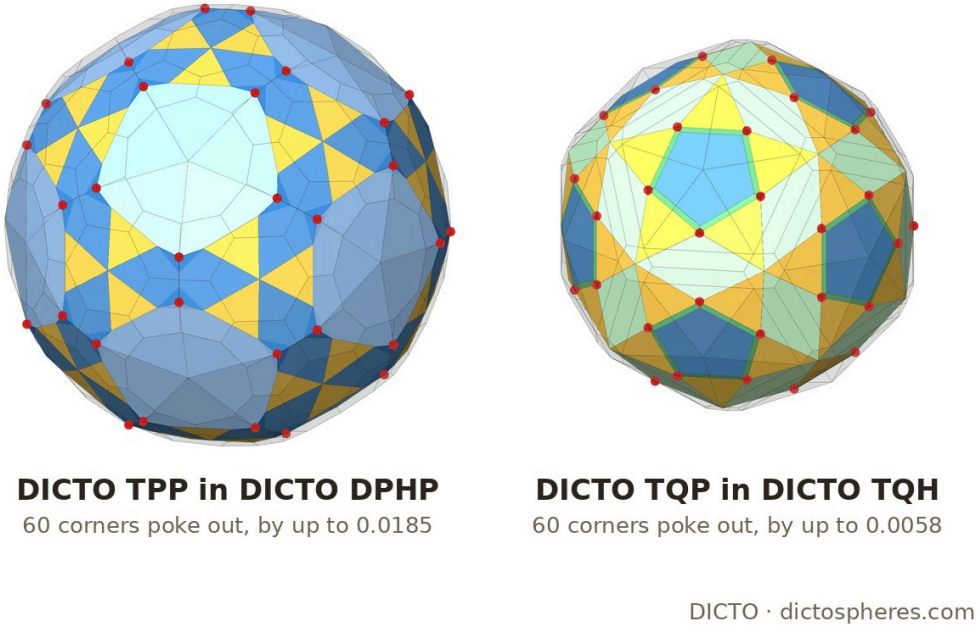


Figure 2. Where the nesting breaks. Left: the DICTO TPP inside the DICTO DPHP (glass); right: the DICTO TQP inside the DICTO TQH (glass). Red: the 60 corners that lie outside (front half shown).

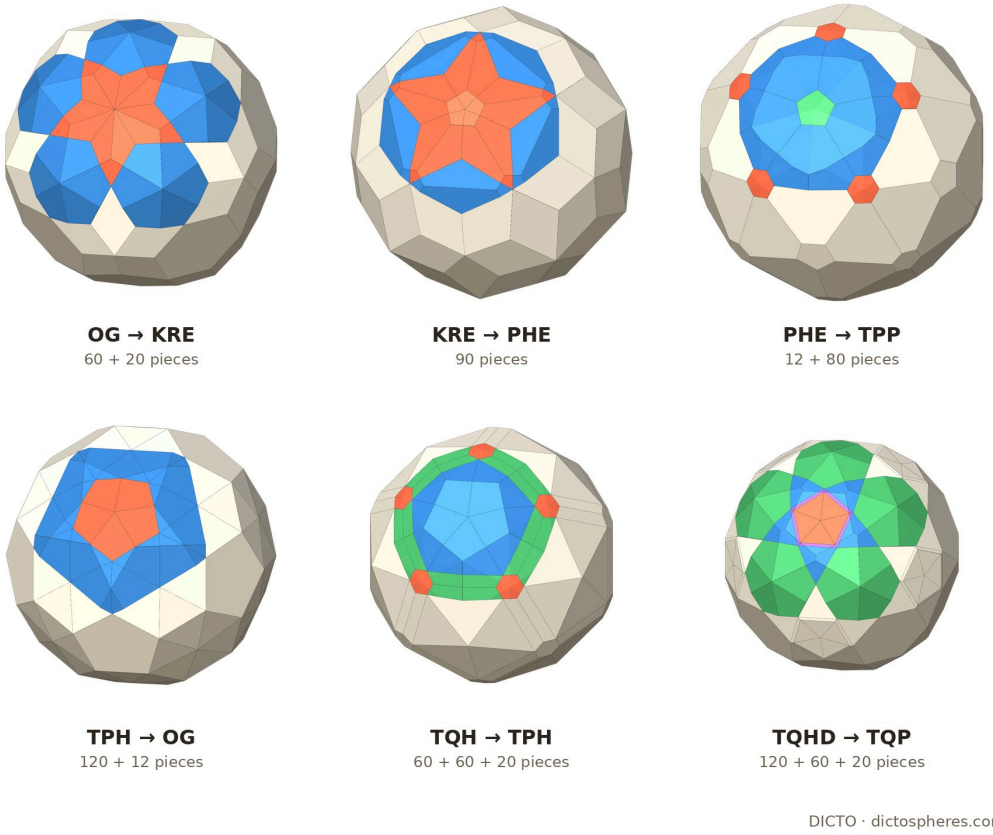


Figure 3. The six nested shells, each filled exactly: top, upward (OG → KRE, KRE → PHE, PHE → TPP); bottom, downward (TPH → OG, TQH → TPH, TQHD → TQP). Each shows the inner solid (pale) with its shell pieces on the faces round one five-fold axis, one colour per kind of piece.

No cycle. From the icosidodecahedron, T returns to the start, scaled by $10/\varphi^4$, after four steps [2]. From the OG neither direction returns within four steps: the corner counts grow both ways (92, 180, 212, 270 up; 80, 132, 140, 162 down), and the face kinds multiply. In the plane, iterated antipedal triangles tend to the equilateral triangle [5]; the DPHP is the roundest of the nine (corner radii within 0.56%, against the OG's 4.28%), but roundness does not grow at every step (the TPP: 5.13%), and we make no claim beyond four steps.

5. Harmonic fingerprint and Dehn invariant (Version 2)

Version 2 reads the nine solids, TQHD, TQP, TQH, TPH, OG, KRE, PHE, TPP and DPHP in the order of the chain, with two tests from other work: their harmonic fingerprint, as DICTO proposed for the golden nests and the OG family, and their Dehn invariant, the Dehn column of The DICTO Concordance. Both are run on the float corners of the exact builds (data/og-family-solids.json). **Numbers marked numerical are float values**; the statements marked exact are proved.

The fingerprint [9, 14]. Give each corner of a solid, seen from its centre, a unit weight on the sphere; with $q_{lm} = (1/N)\,\Sigma\, Y_{lm}(u_i)$ over the N corner directions u_i, $Q_l = (4\pi/(2l+1)\,\Sigma_m\, |q_{lm}|^2)^{1/2}$, and $\hat{W}_6$ is the normalised third-order invariant of degree 6 built with Wigner 3j symbols [9]. $\hat{W}_6$ separates the dual pair: -0.16975 for the icosahedron (corners on the five-fold axes), $+0.16975$ for the dodecahedron (corners on the three-fold axes).

> **Proposition 1** (fingerprint of the OG family). (a) *Exact.* Each of the nine solids has $Q_4 = Q_8 = 0$, and $\hat{W}_6 = \pm 0.16975$, the value of the icosahedron up to sign. (b) The sign splits the family: $\hat{W}_6 < 0$, of corner type like the icosahedron, for the TQHD, TQP, OG and KRE; $\hat{W}_6 > 0$, of face type like the dodecahedron, for the TQH, TPH, PHE, TPP and DPHP. Along the chain TQHD, TQP, TQH, TPH, OG, KRE, PHE, TPP, DPHP the signs read $- - + + - - + + +$: they change in pairs, not at every step. (c) *Numerical.* The KRE is the quietest of the nine: it has the smallest Q_{10} (0.0103) and Q_{12} (0.0038), and the smallest $(Q_6^2 + Q_{10}^2 + Q_{12}^2)^{1/2}$, 0.042, against 0.108 for the next two (TPH and DPHP). The smallest Q_6 is the TQP's, 0.032.

Proof of (a). The nine solids have the full icosahedral symmetry (§§2–3), so q_{lm} is the projection of the corner weights onto the icosahedral invariants of degree l. The icosahedral group has no invariant spherical harmonic of degree 4 or 8, and exactly one, up to scale, of degree 6 (Klein [10]): so $Q_4 = Q_8 = 0$, and the degree-6 part of every icosahedral set is a real multiple c of one fixed harmonic. $\hat{W}_6$ is cubic in q_{6m} and divided by the cube of its norm, so it equals the icosahedron's value times the sign of c whenever $Q_6 \neq 0$. Only the sign of $\hat{W}_6$ carries information in this family. □ Parts (b) and (c) are float values (method calibrated on the icosahedron, $Q_6 = 0.6633$, the classical value [9]); the signs of (b) are far from the rounding level, since $|\hat{W}_6| = 0.16975$ for all nine and $Q_6 \geq 0.032$.

The sign is not the alternation of corners and faces on the axes from step to step: it records which of the two icosahedral shapes of degree 6, the icosahedron's or the dodecahedron's, outweighs the other in the corner set.

> **Proposition 2** (Dehn invariant). All nine solids have a nonzero Dehn invariant. For the OG this is exact (#96, Reading the Voids [15], Theorem 1); for the other eight it is numerical, a calibrated screen. So none of the nine fills space alone [12], and none can be cut into finitely many pieces and reassembled into a cube [11].

Method (numerical). $\mathrm{Dehn}(S) = \Sigma_e L_e \otimes \theta_e$ over the edges, with length L_e and dihedral angle θ_e [11]. Integer relations among the distinct dihedral angles and π are found by PSLQ [13] (coefficients up to 300, tolerance 10^{-8}); the angles that stay are a basis independent modulo $\mathbb{Q}\pi$, and $\mathrm{Dehn}(S) = 0$ exactly when, for each basis angle, the edge lengths weighted by its coefficients sum to zero. Calibration on ten solids with a known answer, all correct: zero for the cube, the rhombic dodecahedron, the truncated octahedron, the rhombic triacontahedron and the tetrahedron with its corners cut off at a quarter of each edge; nonzero for the tetrahedron, the octahedron, the icosahedron, the dodecahedron and the Archimedean truncated tetrahedron. Every solid of the family has two basis angles, and its weighted sums are between 26 and 1476 times its longest edge (Table 4), where a zero shows below 10^{-9}; the result is the same at tolerance 10^{-9}. The rhombic triacontahedron shows that the result is not automatic for icosahedral solids: its one dihedral angle is $144° = 4\pi/5$, and its Dehn invariant is zero. An exact proof for the eight needs their exact dihedral angles.

Solid	$\hat{W}_6$ sign	Q_6	Q_{10}	Q_{12}	Dihedral angles	Basis angles	Smallest |sum|
DICTO TQHD	−	0.1867	0.0674	0.0639	5	2	398.8
DICTO TQP	−	0.0322	0.1028	0.1213	4	2	44.2
DICTO TQH	+	0.2330	0.3846	0.0428	4	2	52.0
DICTO TPH	+	0.0529	0.0426	0.0834	3	2	139.6
DICTO OG	−	0.1327	0.0934	0.0265	2	2	49.7 (exact: nonzero)
DICTO KRE	−	0.0405	0.0103	0.0038	2	2	49.8
DICTO PHE	+	0.0388	0.3206	0.0601	3	2	26.2
DICTO TPP	+	0.0933	0.1157	0.0848	5	2	57.2
DICTO DPHP	+	0.0370	0.0847	0.0564	4	2	40.6

Table 4 (numerical). Fingerprint ($Q_4 = Q_8 = 0$ for all nine, exact) and Dehn screen of the nine solids. Dihedral angles: the number of distinct angles at the edges; basis angles: those independent modulo $\mathbb{Q}\pi$; smallest |sum|: the smaller of the two weighted edge-length sums, in units of the solid's longest edge (zero would be below 10^{-9}). For comparison: icosahedron Q_6 0.6633, dodecahedron 0.3685, icosidodecahedron 0.2073 (same method). Studies 2026-11-04-nest-fingerprint (og-fingerprint.json) and 2026-11-04-dehn-screen (data/dehn-screen.json).

6. Prior art and what is claimed

- **The steps.** In the plane, T and P are the classical *antipedal* and *pedal* constructions about a point: the antipedal polygon has its sides through the vertices, at right angles to the lines from the point, and the pedal polygon has its vertices at the feet of the perpendiculars to the sides [3, 4, 5]. Iterated pedal and antipedal triangles have long been studied [5]. For polyhedra Smarandache [3] proposed the “face pedal polyhedron” (feet of the perpendiculars from an interior point to the faces), which is P, and posed its properties as an open problem; we found no study of it or of the 3-D antipedal step T. **This sharpens [2]**, which reported no source for T: T is the 3-D form of the antipedal construction about the centre.
- **The KRE** is of a known type: the rhombic enneacontahedron has 92 corners and 90 rhombi, 60 wide and 30 narrow, and “without the equal edge constraint, the wide rhombi are kites if limited only by the icosahedral symmetry” [8]. The KRE is such a solid; what is new is its shape, fixed exactly by the OG.
- **The PHE** has the counts of the Goldberg polyhedron GP(3,0) [6, 7]; its regular pentagons and 20 regular hexagons with 60 irregular ones, fixed by the OG, we found nowhere.
- **Version 2 (§5).** The fingerprint Q_l, $\hat{W}_6$ is Steinhardt, Nelson and Ronchetti's [9], used as in #94 [14]; the icosahedral invariants of degrees 6, 10 and 15 are Klein's [10]; the Dehn invariant and its theorem are Dehn's [11], the zero Dehn invariant of every space-filler is Debrunner's [12], and the OG's nonzero invariant is proved in #96 [15]. The other eight solids appear in no source (below), so neither do their fingerprints or Dehn invariants.
- **The others.** We found no polyhedron with the counts of the TPP (60 triangles + 120 pentagons, 212 corners), the DPHP (12 decagons + 20 hexagons + 120 pentagons, 270 corners), the TPH (120 triangles + 12 pentagons, 80 corners), the TQH, the TQP or the TQHD.

In the sources checked (the full text of the 80 library sources listed in the appendix, and web searches on 20 October 2026) we did not find the eight solids, their exact shapes, the shells and pieces, or the two breaks. **Claimed:** the eight solids as the exact steps of T and P from the OG, their exact faces and volumes, the exact shells and their pieces, and where and why the nesting breaks; in Version 2, the exact part (a) of Proposition 1, and as numerical results its parts (b) and (c) and Proposition 2 for the eight solids other than the OG. **Not claimed:** the steps themselves (antipedal and pedal), the KRE's type, or anything beyond four steps.

7. Further work

- Short papers on single solids of the family, after this one.
- The family in the Kaleidohedra app, with the breaks shown as a feature (the TPP's corners through the DPHP).
- Exact dihedral angles of the eight solids other than the OG, to make Proposition 2 exact for them.
- Steps beyond four, which need exact arithmetic: float cannot be trusted there, as nearly coplanar faces merge or split.

Data and reproducibility

Exact scripts ($\mathbb{Q}(\sqrt{5})$), in studies/2026-10-19-five-fold-octet at the commit above: exact_upward.py (KRE, PHE, TPP, DPHP and their shells), exact_downward.py (TPH, TQH, TQP, TQHD, T(P(S)) = S, nesting and shells), family_facts.py (the OG in this frame, P(KRE), regular faces, the OG's five-fold corners), built on exact_cycle.py and exact_shells.py from [2]. Logs and data: exact-downward.log, family-facts.log, data/exact_upward.json, data/exact_downward.json, data/family_facts.json, data/og-family-solids.json (float corners and ordered faces of all nine solids). Figures: family_figs.py, shots.mjs, family_compose.py.

Version 2 (§5, numerical apart from Proposition 1(a)). Fingerprint: studies/2026-11-04-nest-fingerprint (README.md, og-fingerprint.json: Q_4, Q_6, Q_8, Q_{10}, Q_{12} and $\hat{W}_6$ of the nine solids and of the icosahedron, dodecahedron and icosidodecahedron), method Q3d and W6 in studies/_lib/methods.py, calibrated in its test_methods.py. Dehn screen: studies/2026-11-04-dehn-screen (scripts/dehn.py, data/dehn-screen.json, README.md: the ten calibration solids and the nine solids). Both read the corners of data/og-family-solids.json of this paper's study.

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15. DICTO, Reading the Voids: Spherical Waves, Harmonics and Golden Bricks in the Gaps of a Packing (#96), DICTO DOI: 10.5281/zenodo.23968272.

Appendix. The reference library searched

The full text of the 80 sources below in DICTO's reference library was searched on 20 October 2026 for: pedal, antipedal, reciprocation, polar, canonical, foot of the perpendicular, enneacontahedron, Goldberg, kite, and the corner and face counts of the eight solids. None contains the results of this paper. The relevant hits are Goldberg's paper and the Wikipedia page on Goldberg polyhedra [6, 7], and Kabai's and Grünbaum's mentions of the rhombic enneacontahedron. Thirteen further items are page images without a text layer (the patents of Fuller, Snelson, Pearce, Baer, Hogan, Bergman and Lalvani, and Fedorov 1885).

Web searches (20 October 2026): pedal and antipedal polyhedra and their iteration; polyhedra with the counts of each of the eight solids; 30 rhombi with 60 kites; iterated polar reciprocation through perpendicular planes.

Author(s)	Year	Title
Luca Pacioli, Leonardo da Vinci (ill.)	1509	Divina proportione
Albrecht Dürer	1525	Underweysung der Messung mit dem Zirckel und Richtscheyt
Wenzel Jamnitzer, Jost Amman (engr.)	1568	Perspectiva corporum regularium
Johannes Kepler	1596	Prodromus dissertationum cosmographicarum continens Mysterium cosmographicum
Johannes Kepler	1619	Harmonices mundi libri V
Euclid, T. L. Heath (tr.)	1908	The Thirteen Books of Euclid's Elements, Vol. III (Books X-XIII)
M. Goldberg	1937	A class of multi-symmetric polyhedra
Haresh Lalvani	1990	Transformational part-count in layered octahedral-tetrahedral truss configurations (NASA TM-102635)
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M. Baake, F. Gähler	1998	Symmetry structure of the Elser-Sloane quasicrystal
T. C. Hales	1998	An overview of the Kepler conjecture
M. Baake	1999	A guide to mathematical quasicrystals
O. Ogievetsky, Z. Papadopolos	1999	On Quasiperiodic Space Tilings, Inflation and Dehn Invariants
U. Betke, M. Henk	1999	Densest lattice packings of 3-polytopes
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Z. Papadopolos, O. Ogievetsky	1999	On quasiperiodic space tilings, inflation and Dehn invariants
R. V. Moody	2000	Model sets: a survey
M. Senechal	2004	Donald and the golden rhombohedra
S. Kabai	2007	54-Faced Space-Filling Polyhedron (Wolfram Demonstration; description and screenshot)
S. Kabai	2007	Golden Octet Truss (Wolfram Demonstration; description and screenshot)
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S. Kabai	2008	A Rhombic Dodecahedron Made of Rhombic Triacontahedra (Wolfram Demonstration; description and screenshot)
S. Kabai	2008	Relating a Rhombic Triacontahedron and a Rhombic Dodecahedron (Wolfram Demonstration; description and screenshot)
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